

Functions in JAVA



“Your future is created by what you do today, not tomorrow.”

WHY DO WE NEED A FUNCTION

To avoid writing repeated lines of code wherever needed.

```
int main() {
```

```
int a = 10, b = 20;
```

```
int c = a + b;
```

```
int d = 2 * c;
```

```
int x = 40, y = 50;
```

```
int z = x + y;
```

```
int d = 2 * z;
```

```
}
```

**Same lines of code with
different value**



WHAT IS FUNCTION

Set of instructions, which mainly performs specific task.

```
return_type function_name(parameters) {  
  
    // body  
  
}
```



TYPES OF FUNCTIONS

Set of instructions, which mainly performs specific task.

No parameters, no return

No parameters, with return

With parameters, no return

With parameters, with return



No parameters, no return

Syntax

```
void function_name() {
```

```
    // body
```

```
}
```

```
void greet() {
```

```
    System.out.println("Codegreedy");
```

```
}
```

```
int main() {
```

```
    greet(); // calling
```

```
}
```

No parameters, With Return Value

Syntax

```
return_type function_name() {
```

```
    // body
```

```
}
```

```
int getNumber() {  
    return 10;  
}
```

```
int main() {  
    int num = getNumber();  
    System.out.println(num);  
}
```

With Parameters, No Return Value

Syntax

```
void function_name(par1, par2) {
```

```
    // body
```

```
}
```

```
void printSum(int a, int b) {  
    System.out.println(a + b);  
}
```

```
int main() {  
    printSum(3, 4);  
}
```

With Parameters, No Return Value

Syntax

```
void function_name(par1, par2) {
```

```
// body
```

```
}
```

```
void printSum(int a, int b) {  
    System.out.println(a + b);  
}
```

```
int main() {  
    printSum(3, 4);  
}
```


With Parameters, With Return Value

Syntax

```
returnType function_name(par1, par2) {  
  
    // body  
  
}
```

```
int add(int a, int b) {  
    return a + b;  
}
```

```
int main() {  
    int result = add(5, 6);  
    System.out.println(result);  
}
```